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Prediction-Driven Dual-Timescale Resource Allocation for Long-Term QoS in Beam-Hopping LEO Satellites

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***Abstract*—Beam-hopping low Earth orbit (LEO) satellite systems support flexible coverage for uneven traffic, but dynamic topology, bursty demand, and limited onboard resources make long-term quality-of-service (QoS) support challenging. Existing methods mainly optimize instantaneous throughput and thus struggle to guarantee long-term service reliability. This paper proposes a prediction-guided, queue-corrective dual-timescale architecture for beam-hopping LEO systems. At the window timescale, a Temporal Fusion Transformer (TFT) predicts traffic demand and guides coarse power budgeting. At the slot timescale, a Lyapunov virtual-queue mechanism enables queue-aware online scheduling and mitigates prediction mismatch. In a representative small-scale single-satellite scenario (7 cells, 14 users), the proposed method improves long-term user satisfaction by 40% over myopic baselines and achieves better cell-edge performance.**

***Index Terms*—Beam-hopping LEO satellites, long-term QoS, resource allocation, spatio-temporal prediction, Lyapunov optimization, rate-splitting multiple access.**

I. Introduction

Beam-hopping low Earth orbit (LEO) satellite systems provide flexible coverage for highly uneven traffic distributions, but their dynamic topology, time-varying channels, and limited onboard resources make long-term QoS provisioning challenging. Existing beam-hopping and RSMA/HRS-based methods mainly optimize instantaneous throughput, beam scheduling, or interference management [1]–[7], and therefore struggle to guarantee long-term service reliability under bursty traffic and channel variations. Recent surveys and related studies on satellite networking, learning-based resource allocation, and RSMA-enabled multibeam systems can be found in [8]–[11].

Existing studies still lack a unified framework that jointly handles slow traffic evolution, fast channel variation, and long-term QoS in beam-hopping LEO systems. Existing prediction-driven methods often lack slot-level corrective feedback when forecasts deviate from actual demand, while queue-aware control approaches react only after service deficits occur. To address this gap, we propose a **prediction-guided, queue-corrective dual-timescale framework** that couples *window-level prediction guidance* with *slot-level queue correction*. The main contributions are: 1) a dual-timescale framework for long-term QoS-oriented resource allocation; 2) a satisfaction-oriented virtual queue that prioritizes deficit users online and mitigates moderate prediction mismatch in practice; and 3) dynamic user grouping and adaptive SIC as supporting mechanisms for improved fairness with polynomial complexity.

II. System Model and Problem Formulation

A. Network Architecture

We consider a downlink LEO satellite communication system employing beam-hopping technology. As shown in Figure 1, the satellite equipped with phased array antennas generates multiple spot beams that can be dynamically activated to serve different cells. The system operates in full frequency reuse mode, where all beams share the same spectrum, leading to co-channel interference that must be carefully managed through resource allocation.

The coverage area is divided into M cells, each containing multiple users. Time is divided into two scales: windows τ (slow timescale, each containing L slots) for resource budgeting, and slots t (fast timescale) for instantaneous scheduling. At the window level, the network management system determines which cells to activate ($a_m(\tau) \in \{0, 1\}$) and allocates power budgets $B_m(\tau)$ satisfying $\sum_m B_m(\tau) \leq LN_{max}$, where N_{max} is the maximum number of simultaneous beams.

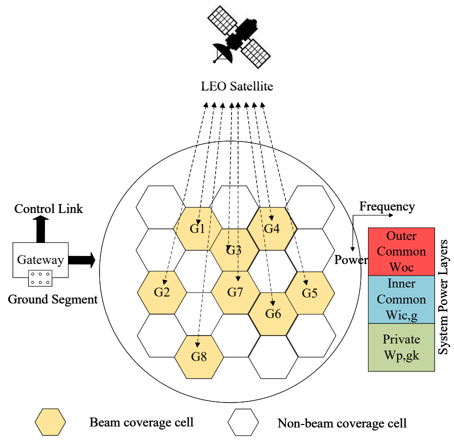


Figure 1. System architecture of beam-hopping LEO satellite network with Hierarchical Rate Splitting (HRS).

B. HRS Transmission Model

At the slot level, we employ Hierarchical Rate Splitting (HRS) transmission to manage multi-user interference. The transmit signal contains three layers: the outer common message s_{oc} for all active users, the inner common messages $s_{ic,g}$ for user groups, and the private messages s_{gk} for individual users. User k's achievable rate $R_k(t)$ depends on the Signal-to-Interference-plus-Noise Ratio (SINR) $\gamma_k(t)$ through

$$R_k(t) = \log_2(1 + \gamma_k(t)), \tag{1}$$

where the SINR accounts for path loss, shadowing, and co-channel interference from simultaneously activated beams under full frequency reuse [5]–[7], [10].

C. Long-Term Satisfaction and Virtual Queues

To quantify long-term QoS, we define the satisfaction indicator for user k at slot t as $S_k(t) = 1$ if the achieved rate meets or exceeds the demand $R_k(t) \geq D_k(t)$, and 0 otherwise. The long-term satisfaction is the time average $\bar{S}_k = \frac{1}{T}\sum_{t=1}^{T} S_k(t)$. This demand-matching metric is more suitable than pure average-rate objectives for bursty and heterogeneous satellite traffic, since it directly reflects whether user service requirements are met over time. Since the binary metric is difficult to optimize directly online, we use a rate-deficit virtual queue as a tractable surrogate for queue-aware control.

Specifically, define the virtual queue $Q_k(t)$ as

$$Q_k(t+1) = \max\{Q_k(t) + D_k(t) - R_k(t), 0\}, \tag{2}$$

where $D_k(t)$ denotes the instantaneous traffic demand of user k. A larger $Q_k(t)$ indicates persistent service deficit and thus higher urgency.

Define the Lyapunov function as

$$L(\mathbf{Q}(t)) = \frac{1}{2}\sum_{k=1}^{K} Q_k^2(t), \tag{3}$$

and the drift-plus-penalty term as

$$\Delta_V(t) = \mathbb{E}\left[L(\mathbf{Q}(t+1)) - L(\mathbf{Q}(t)) - V\sum_{k=1}^{K}\omega_k R_k(t) \mid \mathbf{Q}(t)\right], \tag{4}$$

where $V > 0$ is the Lyapunov control parameter and ω_k is a fixed priority weight. By standard Lyapunov drift analysis, minimizing an upper bound of $\Delta_V(t)$ at each slot is equivalent, after dropping terms independent of the slot decision, to maximizing the queue-aware weighted sum-rate:

$$\begin{aligned} \max_{\chi(t)} \quad & \sum_{k=1}^{K} W_k(t) R_k(t) \\ \text{s.t.} \quad & \sum_{m=1}^{M} x_m(t) \leq N_{max}, \\ & \sum_{m} P_m(t) \leq P_{max}, \\ & \text{HRS decodability constraints,} \end{aligned} \tag{5}$$

where $W_k(t) = V\omega_k + Q_k(t)$ and $\chi(t)$ collects the slot-level decision variables, including beam activation $x_m(t)$, power allocation $P_m(t)$, and precoding variables.

III. Proposed Dual-Timescale Algorithm

A. Window-Level Planning with TFT Prediction

At the window level, we employ Temporal Fusion Transformer (TFT) [12] to predict spatio-temporal traffic demands $\hat{D}_m(\tau)$ and channel efficiency $\eta_m(\tau)$ for each cell. TFT effectively captures both temporal patterns and spatial correlations between cells. The role of this slow-timescale planner is not to determine exact slot decisions, but to provide coarse-grained resource guidance robust to traffic fluctuations.

Given predictions, we formulate the window planning problem to maximize weighted satisfaction via capped utility:

$$\max_{\{B_m(\tau)\}} \sum_{m=1}^{M} \omega_m \min\left\{1, \frac{\eta_m(\tau) B_m(\tau)}{L\hat{D}_m(\tau)}\right\}, \tag{6}$$

subject to $\sum_m B_m(\tau) \leq LN_{max}$ and $0 \leq B_m(\tau) \leq L$. The capped utility equals one once the predicted window demand is fully met, and otherwise increases linearly with the allocated budget. Since it is monotone and piecewise concave in $B_m(\tau)$, the relaxed planning problem is a convex program and can be efficiently solved in $O(M \log M)$ by a sorting/water-filling procedure.

B. Slot-Level Optimization via Fractional Programming

At each slot, given the budget constraints from window planning, we solve the weighted sum-rate maximization using Fractional Programming (FP) [13]. The non-convexity arises from the coupled log-SINR rate terms under interference and HRS decodability constraints. Using a standard FP-based reformulation, the problem is converted into an iterative procedure that alternates between updating auxiliary SINR-related variables and solving a convexified power-allocation subproblem. This yields a stationary point with complexity $O(N_{iter}K^3)$ per slot, where $N_{iter} \leq 3$ typically suffices. More importantly, the Lyapunov queue weights naturally mitigate prediction mismatch, since persistently underserved users automatically obtain higher priority in subsequent slots.

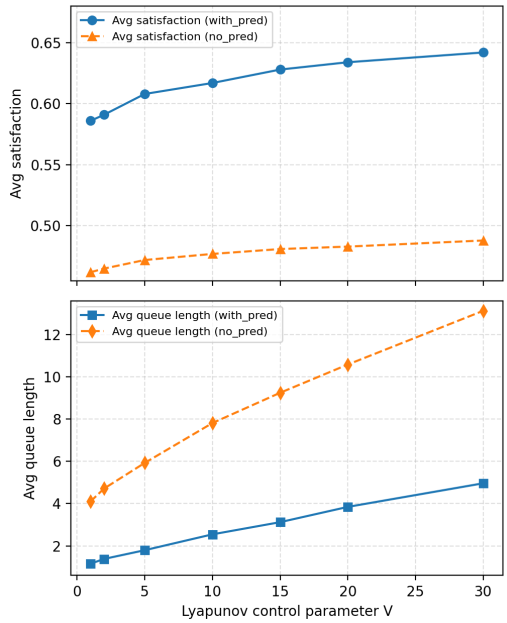


Figure 2. Performance-delay tradeoff versus Lyapunov parameter V under prediction-guided and no-prediction settings.

C. Dynamic User Grouping based on Statistical CSI

To adapt to channel variations at the window timescale, we propose dynamic user grouping based on statistical CSI. Since statistical CSI evolves more slowly than instantaneous SINR, grouping is updated at the slow timescale. Define the spatial similarity between users i and j using the cosine similarity of average channel vectors:

$$s_{i,j} = \frac{|\bar{\mathbf{h}}_i^H \bar{\mathbf{h}}_j|}{\|\bar{\mathbf{h}}_i\|\|\bar{\mathbf{h}}_j\|}, \tag{7}$$

where $\bar{\mathbf{h}}_i$ is the time-averaged channel vector over the window.

The grouping utility function balances intra-group similarity against inter-group interference:

$$U_{\text{grp}} = \sum_{g=1}^{G} \frac{1}{|G_g|} \sum_{i,j \in G_g} s_{i,j} - \lambda_{\text{inter}} \sum_{g<g'} \frac{1}{|G_g||G_{g'}|} \sum_{i \in G_g} \sum_{j \in G_{g'}} s_{i,j}. \tag{8}$$

We employ K-means clustering for initialization, followed by local search where users are iteratively migrated between groups if utility improves. This executes every T_{grp} windows with complexity $O(K^2 + KGT_0)$.

D. Dynamic SIC Decoding Order

At each slot, given power allocation from FP optimization, we dynamically adjust the Successive Interference Cancellation (SIC) decoding order based on instantaneous weights and SINR conditions. Since decoding priority depends on instantaneous interference and queue urgency, SIC ordering is updated at the fast timescale.

Define the importance metric for user k's private layer as

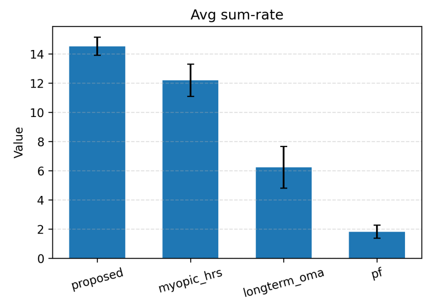
$$\pi_k^{pri}(t) = W_k(t) \cdot \log_2(1 + \gamma_k^{pri}(t)), \tag{9}$$

where $W_k(t)$ is the Lyapunov weight and $\gamma_k^{pri}(t)$ is the instantaneous SINR. For common layers, we use the bottleneck principle:

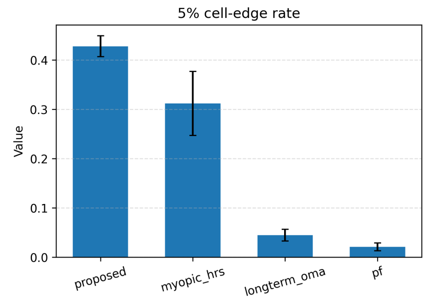
$$\pi_{c,g}(t) = \min_{k \in G_g} \{W_k(t) \cdot \log_2(1 + \gamma_{k,c,g}(t))\}, \tag{10}$$

ensuring that the worst user in the group can decode the common message.

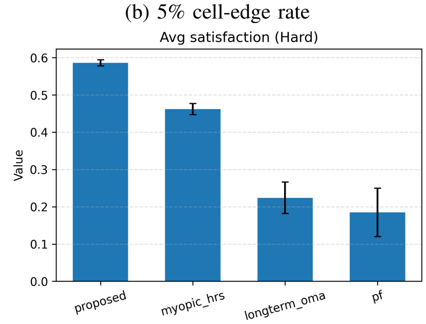
The decoding order follows descending π values, so high-priority users are decoded earlier and their interference can be removed before subsequent layers. This improves weighted service reliability without re-optimizing power.



(a) Avg sum-rate



(b) 5% cell-edge rate



(c) Hard satisfaction

Figure 3. Overall performance comparison.

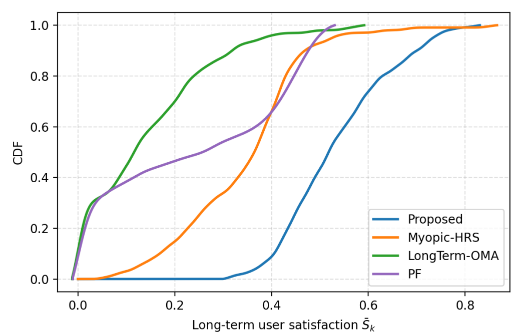


Figure 4. CDF of long-term user satisfaction.

IV. Complexity and Convergence Analysis

The proposed framework operates hierarchically at three timescales. Window-level prediction and budget allocation require $O(M \log M)$, while slot-level FP optimization requires $O(N_{iter}K^3)$ with typically small N_{iter}. Dynamic grouping incurs $O(K^2+KGT_0)$ and is amortized over multiple windows. The slot-level FP optimization converges to a KKT stationary point because each iteration monotonically improves an upper-bounded objective. Under standard bounded-arrival and feasible-control assumptions, Lyapunov theory ensures strong stability of the virtual queues, which implies long-term QoS satisfaction up to the usual $O(1/V)$ trade-off.

V. Simulation Results and Discussion

We evaluate the proposed framework in a representative small-scale single-satellite scenario with 7 cells, 14 users, and 800 slots. Accordingly, the results and conclusions in this paper are restricted to the considered setting. We compare four schemes: Proposed, Myopic-HRS, LongTerm-OMA, and PF. The Lyapunov parameter V is selected from a moderate range to balance satisfaction improvement and queue growth. Fig. 2 illustrates this performance-delay tradeoff under prediction-guided and no-prediction settings.

Figure 3 presents the comprehensive performance comparison. The proposed method achieves comparable average sum-rate to Myopic-HRS while significantly improving the 5% cell-edge rate by 35%. Most importantly, the hard satisfaction metric shows 40% improvement over Myopic-HRS, demonstrating the effectiveness of long-term QoS optimization. These results indicate that the proposed framework improves long-term service reliability without paying a prohibitive throughput penalty.

Figure 2 examines the sensitivity to the Lyapunov control parameter V. As V increases, average satisfaction exhibits diminishing returns, while queue length grows steadily, reflecting the classical performance-delay tradeoff. The comparison between prediction-guided and no-prediction settings further highlights the value of prediction guidance, while showing that the queue-correction mechanism still maintains reasonable service quality in the considered setting.

Figure 4 shows that 95% of users achieve satisfaction > 0.6 under the proposed scheme, whereas the baselines remain substantially lower, confirming improved long-term fairness in the considered setting.

VI. Conclusion

We proposed a prediction-guided dual-timescale resource allocation framework for beam-hopping LEO satellite systems, improving long-term QoS and fairness with competitive efficiency in the considered setting. The results highlight the value of coupling window-level prediction guidance with slot-level queue correction for long-term service reliability.

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